

Anudeep Peela

Bangalore, India | anudeeppeela9@gmail.com | +91-93813-42465 | anudeep-peela.github.io
linkedin.com/in/anudeep-peela | github.com/anudeep-peela

Senior Data Scientist | Machine Learning, Optimization & Production AI

Professional Summary

- Senior Data Scientist with **4 years** at McKinsey building applied ML, optimization, NLP, and production AI systems across banking, manufacturing, pharma, healthcare, mobility, and automotive domains.
- Delivered pricing, workforce-allocation, steam-loss, voice-analytics, and GenAI workflows with measurable outcomes: **20%+** model lift, **45%** utilization gain, **90%** faster analysis, and approved revenue / savings opportunities.

Skills Matrix

ML & Analytics: XGBoost, Scikit-learn, SHAP, regression, segmentation, intent classification, model evaluation

Optimization & OR: PuLP, constrained optimization, pricing grids, workforce planning, genetic algorithms, KNN

Data & Production: Python, SQL, PySpark, pandas, Databricks Jobs/Pipelines, PowerBI, Azure

GenAI & NLP: Azure OpenAI, OpenAI Agents SDK, LangChain, RAG, hybrid retrieval, Pydantic

Experience

McKinsey & Company

Bangalore, India

Senior Data Scientist I

Jan 2026 – Present

- Demand-pattern staffing optimizer:** Optimized staffing across **5** car-rental locations by converting hourly pickup/return demand and weekly patterns into **PuLP**-generated rosters for 3 roles; improved utilization by **45%** while maintaining 5-minute wait-time coverage.
- Underwriting document extraction:** Built a QA/pre-prod extraction pipeline using **Azure OpenAI, OpenAI Agents SDK, Databricks Jobs/Pipelines**, and **Pydantic** outputs; projected to free **3,000+** review hours by extracting loss runs, SOVs, file types, targets, and broker asks with confidence scoring and citations.

Data Scientist II

May 2024 – Dec 2025

- Retail pricing-grid optimization:** Developed an elasticity-based pricing decision engine for Personal and Auto Finance using **Python, SQL, PySpark, and Scikit-learn**; optimized rate grids under 100 bps IRR and maximum 5% sales-decline guardrails, supporting an approved nine-figure SAR opportunity.
- Furnace-level steam optimization:** Deployed a furnace-level **XGBoost** recommendation model across **7** furnaces and **2,000+** sensor tags; combined SHAP, accumulated local effects, and KNN best-performance benchmarking to beat the Advanced Process Control baseline by **20%+**.
- Investigation AI co-pilot:** Deployed a **LangChain**-based RAG co-pilot across **3** pharma sites using hybrid retrieval over historical deviations and CAPA records; reduced historical-analysis time by **90%** and achieved **96%** SME-rated relevance for human-reviewed investigation support.

Data Scientist I

Jul 2022 – Apr 2024

- Voice analytics diagnostic:** Built a GenAI/NLP pipeline over **22K** contact-center transcripts using **LangChain, Azure OpenAI**, and **Pydantic**; achieved **96%** SME-labeled intent accuracy and informed IVR / digital self-service initiatives that drove **30%** call-volume reduction.
- Automotive layout optimizer:** Implemented a genetic-algorithm layout optimizer for automotive engine components, modeling collision, package-volume, pipe-length, and pressure constraints across **7** parts; reduced physical-build testing cycles from about **2 months** to **40 minutes** of virtual testing.

Education & Training

Indian Institute of Technology Bombay

Jul 2017 – Jun 2022

B.Tech. & M.Tech. (Dual Degree), Specialization in Artificial Intelligence

CGPA: 8.0/10

Training: AWS Machine Learning Specialty coursework; MLOps Engineering on AWS